

CLIMB at VBS 2027: Multi-Signal Keyframe Retrieval with Collaborative Ad-Hoc Search

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Abstract. We present CLIMB, an interactive video retrieval system built for the Video Browser Showdown 2027. Because a single visual embedding often lacks the precision needed to capture cues, such as scene text, signage, or specific spoken dialogue, CLIMB retrieves information across four complementary signals. Visual embeddings, on-screen text, generated captions, and speech transcripts, which then are fused by rank rather than score and a solid mix between lexical and semantic matching depending on the signal used, ensure broad retrieval coverage. Queries can be chained into temporal sequences, with each stage represented as an independent query, providing increased flexibility while reducing the search space.

For ad-hoc video search, CLIMB treats a task as a single shared team activity, hiding the shots a teammate has submitted from everybody else’s results, while search, browsing, and submission continue to work, even when the shared service is unreachable.

Keywords: Video Browser Showdown · Interactive Video Retrieval · Collaborative Search · Semantic Embedding.

1 Introduction

The Video Browser Showdown [9] is an interactive video retrieval competition asking participants to solve known-item search (KIS), ad-hoc video search (AVS), and visual question-answering (VQA/QAS) tasks against large scale video collections. For VBS 2027, the main collection is V3C [10] with 28,450 videos, roughly 3,800 hours, and about 4.7 TB, segmented into 4.14 million master shots, alongside the smaller MVK [12] and GynSurgLHE [6] collections.

Two problems follow from the task types. The first is that a single visual embedding is vague when it comes to places where a task hint might be precise. Textual hints can name signs seen on screen or words spoken in a dialog. The second is that AVS scoring rewards distinct correct shots, so a team of n operators searching independently spends a substantial share of its submissions re-finding what a teammate has already sent.

CLIMB addresses both. It indexes every video as scenes and keyframes, retrieves over four independent signals, fuses them by rank, and treats an AVS task as a single shared activity rather than n parallel ones, vanishing duplicate scenes from everybody’s list. Concretely, we contribute:

1. Keyframe-level retrieval fusing four heterogeneous signals by rank, with the choice between lexical and semantic matching made per signal by measurement (Sect. 4.1).
2. A temporal query language, $A \gg B \gg (120) C$, in which every stage is itself a full query (Sect. 4.3).
3. A collaborative AVS mode built around the scoring rule, which degrades safely when the shared service is unreachable (Sect. 6).
4. An index engineered for everyday hardware: binary-quantized HNSW with oversample-and-rerank, fed by a resumable ingest pipeline (Sect. 3).

2 System Overview

CLIMB is built from six components (Fig. 1). Decoding each video, segmenting it into scenes, selecting the keyframes, computing the embeddings, reading the text on screen, as well as captioning and transcribing, happens in an offline pipeline described in Sect. 3. The resulting scenes, keyframes, and four indexes that get queried by the retrievers land in a PostgreSQL database.

At query time the retrieval stack and main application run as separate processes, due to the simple reason that they are written in different languages. While the models and the fusion require a Python environment, media serving, DRES communication, and the AVS mirror are based on Node.js and Express. Both processes connect to the database: the backend reads the scene and keyframe metadata directly and only delegates embedding, ranking, and fusion to the search engine. The React frontend (Sect. 5), however, only talks to the backend. An unavailable search or collaboration service still allows for browsing and submission.

For scenarios where network reliability is a concern, CLIMB supports fully localized deployments where an entire instance runs per operator machine. In this mode, media assets are served directly from localhost, and the shared AVS session service (Sect. 6) accounts for the only network overhead, a few tens of kilobytes of JSON polled every four seconds (amounting to single-digit kilobytes per second per machine).

3 Indexing Pipeline

3.1 Scenes and Keyframe Selection

The V3C dataset provides scene boundaries. These scene boundaries are the center of our encoding. They get used, splitting the video into the different scenes. For each of these scenes, we select between 2 and 32 keyframes according to the length of the scene. According to the following formula:

$$k(s \in S) = \min(\max(\text{ceil}(t(s)/4), 2), 32), \quad (1)$$

where S is the set of scenes in a video, and $t(s)$ is the duration of the scene s in seconds, selecting one keyframe per four seconds.

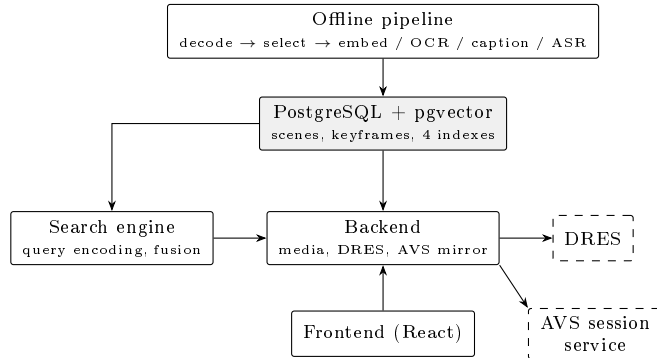


Fig. 1. CLIMB architecture. Every component shown runs locally on each machine, except for the two external services: DRES and the AVS session service, which is deployed once.

We start by selecting the most typical frame and then repeatedly add the frame, which is the most different from the selected keyframes so far. This selects a broad coverage of keyframes in a scene and not near duplicates. Afterward we reject blank frames. A blank frame is a frame that is nearly flat and has almost no edges. This reduced the number of keyframes sampled from our subset (Sect. 7) from 99,661 keyframes at 1 fps to 35,558 keyframes.

Damaged bitstreams are indexed from what we were able to decode, flagged, and logged.

3.2 Feature Extraction

CLIMB extracts four signals per scene. Those were chosen so that each one covers a failure of the others. Where an embedding is vague, a sign or a spoken name is precise. Where there is no visual hint, there might be a spoken one, and where there is no exact text known to search for, there might be a general description that can be indexed.

Whilst we have four different signals, they are deliberately not applied equally. OCR reads every keyframe, but captioning is only done once per scene. The center of the application is SigLIP2, which embeds each keyframe into a 1024-dimensional vector space and handles anything visual [13]. OCR also looks at every keyframe, extracting the text on screen. The model used is PP-OCR [2] via PaddleOCR where available and rapidocr-onnxruntime otherwise.

The caption model only captions the first keyframe of each scene, since it is expensive to caption an image and the caption barely changes over the span of a scene. This results in just 14,345 captions in comparison to 35,558 OCR passes and SigLIP2 embeddings. The model that we intend to use for the large dataset will be Qwen2-VL-2B-Instruct [15], but for testing purposes we used a smaller model (SmolVLM-256M-Instruct [5]) as a first draft.

Additionally, we use Whisper [7] as an ASR model, which transcribes what was said during the video and gets later mapped to scenes via overlapping the timestamps of the transcript with the timestamps of the scenes.

In order to encode the textual descriptions of the caption model as well as the transcripts into semantic embeddings, we use multilingual-e5-base [14], which embeds the texts into a 768-dimensional vector space. For efficient retrieval, all generated embeddings (visual and textual) are stored using an HNSW index [4] over binary-quantized vectors. To maintain search accuracy while minimizing memory overhead, initial search candidates are reranked against their full-precision vectors. This quantization significantly reduces the memory footprint, requiring only 5.3,GB for 12.4M keyframes compared to 28,GB for half-precision storage.

4 Retrieval Model

4.1 Rank-Based Fusion at Keyframe Level

As discussed before, we use four different encoders to encode as broad a spectrum of information as possible. Each signal then returns a list of ranked objects, scenes, or keyframes, respectively. Whilst the visual signal and OCR signal natively return a list of ranked keyframes, we only captioned the first keyframe of each scene. Therefore, its signal is scene-based. In order to not inflate the rank of the first keyframe of each scene, this result is then shared with all keyframes of the scene. For the audio signal, it encodes scenes via segments. Each segment gets a den mapped to scenes, and the rank of a scene gets expanded to the keyframes, just as with the caption encoder.

The keyframes are then merged via weighted Reciprocal Rank Fusion (RRF) [1]. Which works just as RRF, but the signals get weighted depending on their source, allowing us to highlight keyframes found via specific encoders, i.e., OCR, where a lexical hit is much more likely to belong to the correct scene than a similar visual embedding. The updated scoring formula is therefore:

$$RRFscore(d \in D) = \sum_{r \in R} \frac{w(r)}{k + r(d)} \quad (2)$$

where D is the set of keyframes, R is the set of encoders, $w(r)$ is the weight of a specific encoder signal, k is a constant set to 60, and $r(d)$ is the rank the encoder r gave the keyframe d [1]. The constant k was chosen by the authors of the original RFF paper and helps to reduce the influence of outliers.

We use the rank to combine the different encoders, since the four signals do not produce comparable scores on their own. A small 20-query test on our subset shows that the best visual result is a score of 0.17 (due to the modality gap described in [3]), whilst the best caption match was a 0.93. This vast difference in scores shows that to map them to a common scale, we would need to calibrate it per retriever, and also update that every time we decide to change a model. Choosing a ranked based fusion approach, we can skip this sensitiv calibration entirely, whilst still allowing for minor tweaking using the assigned signal weights.

Table 1. Query syntax: a plain query searches all four signals; an operator prefix restricts it to one. >> orders two to four queries into a chain within a time window, confined to a single video.

Input	Effect
a drawing of people on a beach	all four signals search
text: state capitol	on-screen text only
said: new york	spoken words only
-video:00191	exclude a video (repeatable)
a video game sunset >> door	one scene, then the other (same video)

4.2 Semantic and Lexical matching

While cross-modal visual retrieval with SigLIP2 is inherently semantic, the remaining text-to-text signals could theoretically rely on lexical matching. However, strict keyword matching on generated captions suffers from severe vocabulary mismatch. For instance, a query for "law enforcement" shares zero lexical overlap with a highly relevant caption describing "a police uniform". In fact, the terms "law" and "enforcement" do not appear once within our extracted 14,345 captions. Consequently, a purely lexical search would return an empty result set rather than just an inaccurate one.

While ASR transcripts are embedded semantically for conceptual and cross-lingual search, users can invoke a "said:" operator for exact lexical matching of specific quotes.

Conversely, OCR operates as a purely lexical signal to pinpoint exact on-screen text provided in task hints. Semantic matching here is redundant, as general concepts are already captured by the visual embedding.

4.3 Query Language and Temporal Chains

Some retrieval tasks require describing not a single scene, but a temporally ordered sequence of scenes, such as a bridge followed by a beach. A plain text query searches all four signals at once, and the operator prefixes in Table 1 restrict it to a single source, whilst >> joins queries into ordered temporal chains.

The chaining makes the prefixes much more powerful. A query `A >> B >> C` is 3 stages, where each one is a complete query on its own. `text: pegasus >> mountain bike race` finds a scene with a text that reads "pegasus", followed by a scene that matches with mountain bike race, independent of a single encoder.

An unannotated chain searches for scenes in the same video that are at maximum 60 seconds apart, annotated with (120) or (d120); it's 120 seconds. The maximum delta is ten minutes, and the maximum amount of stags in a chain is four.

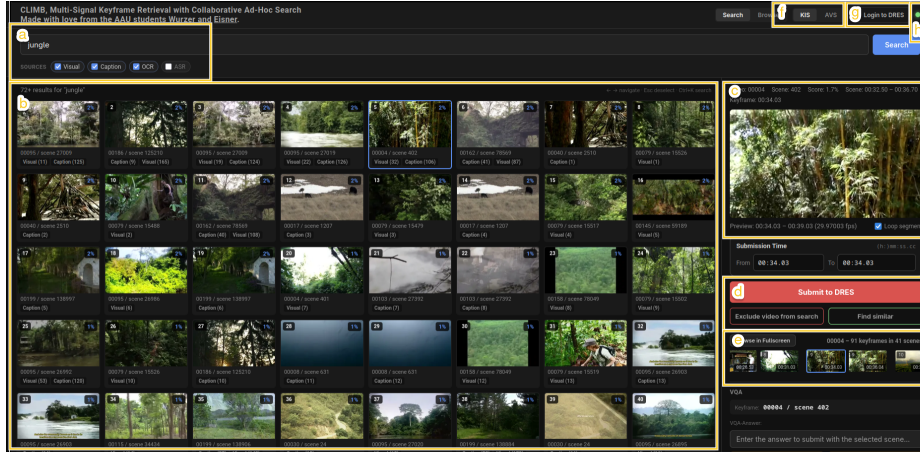


Fig. 2. The CLIMB interface: (a) query bar and source filters, (b) scene result grid, (c) detail panel and video player, (d) submission and search-refinement controls, (e) keyframe filmstrip, (f) AVS session tab, (g) DRES connection controls, and (h) system health status.

5 Interactive Search

5.1 The Search Loop

A search typically follows an iterative interaction loop. The operator enters a query, which returns a grid of keyframes. Selecting one, displays a five-second scene preview starting at the corresponding timestamp. The filmstrip located under the video player, provides an overview of all the keyframes and allows for efficient navigation between them. Keyframe times get populated automatically but remain editable, enabling the operator to adjust the time to the exact frame he wants to submit. Additionally, there is a variety of button based controls support further refinement of the search, completing the search loop.

The large red *Submit to DRES* button submits the selected keyframe. If the keyframe is determined to stem from an irrelevant video, the *Exclude video from search* button removes all keyframes from the search results. The *Find Similar* button on the other hand performs a similarity search using the embedding of the selected keyframe, providing an alternative retrieval strategy when the characteristics of a relevant keyframe are difficult to express.

Further down, the VQA segment, provides the possibility to answer VQA tasks.

To reduce the cost of repeated searches the results are cached and paginated to decrease load times. The initial page triggers the search, whilst later pages are loaded from the cache on scrolling. This allows the operator to inspect additional result pages without reformulating the query when the desired video is not present among the initial results.

5.2 Submitting to DRES

As with every browser at VBS, the DRES-Submission pipeline is an integral part of CLIMB [8,11]. A login dialog can be opened, on the top right where the *server URL*, *evaluation name*, and *credentials* can be entered. If the specific evaluation does not exist, CLIMB falls back to the first available evaluation and explicitly indicates this to the operator.

Every submission returns a status, consisting of a status code, summary, and verdict. This gets displayed to the user after the *Submit to DRES* button. Submissions with a correct verdict are indicated in green, submissions for which no verdict is yet available in yellow, and incorrect or rejected submissions in red. In addition to the default KIS search loop described in Sect. 5.1, the user can submit a text or a shot and text answer to a VQA task.

AVS submissions follow a different processing path, as described in Sect. 6. From the operator’s perspective, however, the interaction is analogous to that of a default KIS task: after switching to the AVS tab in the top-right corner of the interface and joining a session, submissions can be made through the same interaction pattern. The primary difference is that submissions with the results *correct* or *accepted-awaiting-verdict* are hidden from other participants in the same session.

6 Collaborative Ad-Hoc Video Search

Since AVS scoring rewards distinct correct shots, independently searching team members tend to identify the same obvious results first, resulting in a large fraction of duplicate submissions.

We address this problem by making an AVS session a shared state. Submitted scenes are consequently hidden from the results of all session participants. Other scenes in these videos get marked as covered, but not hidden to promote a diverse submission set, whilst allowing for submission of multiple distinct scenes from the same video. An AVS session is identified by a random 4-letter code, whereby each member can create a new session or join one. Sessions expire after 2 hours of inactivity, a duration sufficient to span a task block while limiting persistent server-side state.

Importantly, search does not depend on the AVS collaboration service. Delays or unavailability of the session service therefore cannot block or delay search operations, and submissions remain independent of the collaboration state. To ensure this separation, CLIMB sends the submission request to the DRES server before attempting to communicate with the, independently deployed, collaboration service.

7 Preliminary Results

The pipeline is implemented for the full V3C collection; the numbers below were measured on the 200-video subset available at the time of writing (25,h, 14,345 shots, 35,558 keyframes).

Scene-adaptive selection produces those 35,558 keyframes against 99,661 for uniform 1 fps sampling and still covers every shot.

Compared with exhaustive exact search over 7,543 keyframes (108 ms), over-sampling 1,000 candidates before exact reranking recovers 15 of the true top 20 in 29 ms, 2,000 candidates recover 18 in 54 ms, and 4,000 all 20 in 91 ms. The remaining discrepancy is due to the binary quantization rather than the index: the ANN stage ranks by Hamming distance over 1-bit vectors. CLIMB defaults to 1,000 candidates, while keeping this parameter configurable.

A far simpler version of CLIMB won our university’s course-internal retrieval competition preceding this submission. However we expect VBS to be considerably more challenging. Query latency has been measured at 7,543 keyframes rather than the about 4M shots of the full collection, and neither MVK nor GynSurg resembles the imagery SigLIP2 was trained on.

8 Conclusion

We presented CLIMB, an interactive video retrieval system for VBS 2027. It fuses four heterogeneous signals, namely visual embeddings, on-screen text, generated captions, and transcripts, at the keyframe level using rank-based fusion, with semantic or lexical matching selected independently for each signal. Operators can further chain multiple queries into temporally ordered sequences.

Its collaborative mode is designed around the AVS scoring rule. Shots a teammate has already submitted disappear from everyone’s results, while search, browsing, and submission remain unaffected when the shared session service is unavailable.

Our next steps are to complete ingestion of the full collection and address the domain gap for MVK and GynSurg, where the general-purpose visual embeddings provided by SigLIP2 are expected to be least effective.

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